Joseph Fatoye

Robotics & Autonomous Systems Researcher | Mechatronics Engineer

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Education

Bowen University, Iwo, Nigeria Nov. 2025

B.Eng. Mechatronics Engineering — First Class Honours *CGPA: 4.89/5.00; Rank: 1/50*

- Best Graduating Student, Department of Mechatronics Engineering.

University of Ilorin, Ilorin, Nigeria Sep. 2019 – Oct. 2022

Mechanical Engineering coursework; transferred to Bowen University *CGPA: 4.21/5.00*

Research Interests

Robotics and Autonomous Systems; Visual and Semantic Perception; SLAM; Computer Vision and Multimodal Learning; Trustworthy and Learning-Enabled Autonomy; Control and Decision-Making under Uncertainty; Edge and Embedded AI

Research Experience

Undergraduate Researcher: GRACE Healthcare Robotics Project 2024 – 2026

Bowen University, Iwo, Nigeria *Supervisor: Dr. Segun Adebayo*

- Led a three-student research team developing a low-cost mobile robotics platform for visual localization, semantic perception, and navigation research in healthcare-like indoor environments.
- Integrated monocular ORB-SLAM3 with a C++17 wrapper and Python-based OCR on a Raspberry Pi 5, using process separation, sparse frame sharing, and a translational-stability gate to schedule semantic perception without modifying the SLAM back-end.
- Demonstrated 102 minutes of continuous end-to-end operation over 74,750 processed camera frames with zero system crashes and peak resident memory below 3 GB on the embedded platform.
- Implemented approximate world-frame semantic landmark logging for clinical signs and investigated limitations arising from monocular scale, asynchronous pose association, fixed-depth projection, and constrained onboard computation.

Research Engineer Intern Jun. 2024 – Oct. 2024

Armed Forces Electrical and Mechanical Engineering Workshop, Abuja, Nigeria

- Developed a teleoperated surveillance mobile robot integrating ESP32-CAM and Arduino hardware, L298N motor control, Firebase RTDB communication, and a React-based operator interface.
- Built an ESP32/MPU6050 displacement-estimation prototype and investigated calibration, filtering, and event-triggered logging approaches for mitigating IMU noise and drift.

Selected Robotics & AI Projects

Embedded Plant Classification — Edge AI & TinyML 2026

- Led a six-member team developing an on-device plant classifier using the XIAO ESP32S3 Sense and Edge Impulse; curated 779 field images, trained MobileNetV2 models across multiple input resolutions, and achieved an 84.5% F1-score before deployment for real-time embedded inference.
- Presented the project at the TinyML4D Academic Network 20th Show and Tell in May 2026.

Object Detection Fruit Sorting with Edge Impulse 2023

- Trained and deployed an embedded computer-vision model for fruit recognition, achieving 90% training performance and 84.62% success during real-world testing.

Sunflower Single-Axis Solar Tracker 2024

- Led a team in designing and prototyping an Arduino-based solar tracking system integrating sensing, actuation, and closed-loop positioning logic.

Publications & Research Presentations

- **J. Fatoye** and S. Adebayo, "Stable Semantic Perception for Embedded Robots," Proc. 6th Int. Conf. on Artificial Intelligence and Robotics (MIRG-ICAIR 2026), pp. 119-126, 2026.
- **J. Fatoye**, "Enhancing Robotics with Cognitive Capabilities," *Proceedings of the AAAI Conference on*

Artificial Intelligence, vol. 38, no. 21, pp. 23738–23739, 2024. doi:10.1609/aaai.v38i21.30547.

- "Embedded Plant Classification on ESP32-S3 Using XIAO ESP32S3 Sense," invited project presentation, TinyML4D Academic Network 20th Show and Tell, May 2026.

Technical Skills

Programming:	C++ (17), Python, JavaScript, MATLAB, Bash
Robotics & Autonomy:	ROS, Foxglove, ORB-SLAM3, visual SLAM, Semantic Mapping, Spatial AI, Kinematics
Computer Vision & ML:	OpenCV, TensorFlow, Edge Impulse, MobileNetV2, Embedded Inference, Optical Character Recognition (OCR), Image Preprocessing
Embedded & Hardware:	Microcontrollers (ESP32, Arduino, XIAO S3), Single-Board Computers (Raspberry Pi 5), Sensor Interfacing (IMUs, Time-of-Flight), Actuator Control
Developer Tools:	Git, GitHub, CMake, Linux/Ubuntu, Docker
Design & Prototyping:	SolidWorks, AutoCAD, 3D Printing, Rapid Mechanical Prototyping
Web Development:	React.js, Next.js, Node.js, Flask, Firebase

Honours, Grants & Recognition

- **AAAI Undergraduate Consortium Scholar** 2024
  Selected as one of 19 undergraduate researchers globally; awarded full international travel and registration funding for AAAI-24 (Vancouver, Canada).
- **CVPR Travel Grant** 2024
  Competitive international travel and registration support award (USD 1,190).
- **ECCV Travel Award** 2024
  International registration and travel support stipend (EUR 100).
- **ACL Travel Support** 2023
  Competitive travel grant for conference attendance (USD 350).

Leadership & Academic Service

- Conducted peer tutorials in Engineering Mathematics, Signals and Systems, and Control for more than 50 engineering students at Bowen University.
- Member: IEEE; IEEE Robotics and Automation Society; Black in Robotics; AAAI; Black in AI.
- Held student leadership and welfare roles at Bowen University, supporting academic feedback, peer coordination, and student community activities.